Proof of Sun's conjecture on coprime adjacent sums (OEIS A228886)

Claude (AI model by Anthropic)
run and submitted by Aleksandr Bersenev

October 8, 2026
Produced autonomously by Claude (Opus 5.5) agents in Claude Code; run and submitted by Aleksandr Bersenev.
Not an official Anthropic publication. Not peer reviewed.

Abstract

For $n \geq 1$ let $a(n)$ be the number of permutations $(i_0, i_1, \ldots, i_n)$ of $\{0, 1, \ldots, n\}$ with $i_0 = 0$ and $i_n = n$ such that all $n+1$ cyclic adjacent sums $i_0 + i_1,\ i_1 + i_2,\ \ldots,\ i_{n-1} + i_n,\ i_n + i_0$ are coprime to both $n-1$ and $n+1$ (OEIS A228886). In 2013 Zhi-Wei Sun conjectured that $a(n) > 0$ for every $n \geq 1$ except $n = 2, 4$, and proved this for odd n. We give explicit permutations for all even $n \geq 6$, which completes the proof: $a(n) > 0$ if and only if $n \notin \{2, 4\}$. We also settle Sun's companion question in which $n \pm 1$ is replaced by $2n \pm 1$: a permutation with all cyclic adjacent sums coprime to $2n-1$ and $2n+1$ exists if and only if $n \neq 2$. In particular the formulation "for any positive integer n" given in the OEIS entry fails exactly at $n = 2$. Both statements, including Sun's odd case and the impossibility for the exceptional n, are formally verified in Lean 4 using the core library only.

Contents

1 Introduction **2**

2 Definitions and notation **3**

3 Admissible sums **3**

4 Proof of Theorem 1.2 **4**
4.1 The exceptional values $n = 2, 4$ 4
4.2 Odd n (Sun's permutations) 4
4.3 Even n (new) 5

5 Proof of Theorem 1.3 **8**

6 Formal verification **9**

7 Limitations and open questions **11**

1 Introduction

Sun's 2013 preprint *Some new problems in additive combinatorics* [4] contains a collection of problems asking for permutations or circular arrangements whose adjacent sums or differences satisfy arithmetic restrictions. Many of them are recorded in the OEIS. One of them is the following.

Conjecture 1.1 (Sun, 2013-09-07; [4, Conjecture 3.9], [1])**.** *For every positive integer $n \neq 2, 4$ there is a permutation $i_0, i_1, \ldots, i_n$ of $0, 1, \ldots, n$ with $i_0 = 0$ and $i_n = n$ such that all the $n+1$ adjacent sums*

$$i_0 + i_1, \quad i_1 + i_2, \quad \ldots, \quad i_{n-1} + i_n, \quad i_n + i_0$$

are coprime to both $n-1$ and $n+1$.

The number of such permutations is OEIS sequence A228886 [1]. Its known values begin $1, 0, 1, 0, 1, 8, 3, 24, 78, 1164, 34, \ldots$ for $n = 1, 2, \ldots$, and the entry now lists terms up to $n = 26$, all of them positive except $a(2) = a(4) = 0$. Equivalently, consider the graph on $\{0, \ldots, n\}$ in which i and j are joined when $i + j$ is coprime to $n^2 - 1$; the conjecture asks for a Hamiltonian cycle of this graph through the edge $\{0, n\}$.

What was known. Sun proved Conjecture 1.1 for all odd n on 2013-09-10, by explicit permutations [4, Remark 3.9(ii)]; see also the comments in [1]. We reproduce his permutations in Section 4.2. The even case was left open:

- the OEIS entry, edited as recently as July 2026, records only the odd case as proved; its attached file lists all permutations for $n \leq 10$, but no general construction for even n;
- in a 2020 lecture [5], Sun lists the conjecture with the remark "I have proved this for any positive odd integer n".

Two related results should not be confused with this one. The analogous problem for permutations of $1, \ldots, n$ whose sums are coprime to n (OEIS A228860 [2]) was proved by H.-Q. Cao and H. Pan in 2013, according to that entry. Recent progress on other permutation problems of Sun, such as the prime circle problem for large n and the Hou–Sun problem on primitive-sum graphs over finite fields [8], or the residues of $\sum_k k^2 \tau(k)^2$ modulo $2n+1$ [7], does not address A228886.

In Remark 3.9(i) of [4], Sun also guesses that $n \pm 1$ in Conjecture 1.1 can be replaced by $2n \pm 1$. The comments of [1] state this companion version as: for any positive integer n there is such a permutation with all sums coprime to both $2n-1$ and $2n+1$.

Results. We prove the even case of Conjecture 1.1 by an explicit construction (Section 4.3). Together with Sun's odd case and a short argument for $n = 2, 4$, this gives the following theorem.

Theorem 1.2. *For $n \geq 1$, a permutation as in Conjecture 1.1 exists if and only if $n \notin \{2, 4\}$. Equivalently,* A228886$(n) > 0$ *if and only if $n \notin \{2, 4\}$.*

Theorem 1.3. *For $n \geq 1$, there is a permutation $i_0, \ldots, i_n$ of $0, \ldots, n$ with $i_0 = 0$, $i_n = n$ and all $n+1$ cyclic adjacent sums coprime to both $2n-1$ and $2n+1$ if and only if $n \neq 2$.*

For $n = 2$ the only candidate is $(0, 1, 2)$, and $1 + 2 = 3 = 2n - 1$. So the OEIS wording of the companion conjecture ("for any positive integer n") is false at $n = 2$ and true for every other n. The wording in [4] inherits the exceptions $n \neq 2, 4$ of Conjecture 1.1 and is therefore true.

Both theorems are formally verified in Lean 4 (Section 6). The verification covers the odd case, the even case, the companion problem and the impossibility at the exceptional values.

Novelty. To our knowledge, as of October 8, 2026, neither the even case of Conjecture 1.1 nor the companion conjecture had been proved. Our search covered:

- the OEIS entry and related entries;
- the latest arXiv version (v9, March 2020) of [4], in which Conjecture 3.9 and Remark 3.9 still record only the odd case, and Sun's 2020 slides [5];
- web and arXiv searches with several phrasings, in English and Chinese;
- the OEIS problem list of the *formal-conjectures* project [9], which does not contain A228886, and the lists of OEIS conjectures reported in 2026 as proved by AI agents [10, 11], which do not mention it either.

We could *not* consult Sun's 2021 Chinese-language book of conjectures [6]. If the even case was settled there or anywhere else, our contribution reduces to the companion result and the formal verification. The constructions are short and elementary. The even case seems to have stayed open for lack of attention rather than difficulty.

2 Definitions and notation

Throughout, $n \geq 1$ is an integer, and $\gcd(s, 0) = s$, so that being coprime to 0 means being 1.

Definition 2.1. Let $A, B \geq 0$ be integers. An (A, B)-*cycle of order* n is a sequence $p = (p(0), p(1), \ldots, p(n))$ such that:

1. p is a permutation of $\{0, 1, \ldots, n\}$;
2. $p(0) = 0$ and $p(n) = n$;
3. every cyclic adjacent sum

$$s_i := p(i) + p(i+1) \quad (0 \leq i \leq n-1), \qquad s_n := p(n) + p(0) = n,$$

is coprime to both A and B.

In this language, Theorem 1.2 concerns $(n-1, n+1)$-cycles and Theorem 1.3 concerns $(2n-1, 2n+1)$-cycles. We call an integer s *admissible* when it is coprime to both moduli. Permutations are written as their value sequences $(p(0), \ldots, p(n))$.

The following triviality is used repeatedly. If d divides a and b, then d divides every integer combination $xa + yb$. In particular $\gcd(a, a+k)$ divides k.

3 Admissible sums

Lemma 3.1 (moduli $n \pm 1$)**.** *Let $n \geq 1$. The following sums are coprime to both $n-1$ and $n+1$:*

1. *1 and n, always;*
2. *$n+3$, if n is even; and $n-3$, if n is even and $n \geq 4$;*
3. *$n+2$, if $n \not\equiv 1 \pmod 3$;*

4. $n-2$ *(for* $n \geq 3$*) and* $2n-1$*, if* $n \not\equiv 2 \pmod 3$.

Proof. (1) For 1 this is clear. For n, note $\gcd(n, n \pm 1) \mid 1$.

(2) If n is even, then $n \pm 1$ and $n \pm 3$ are odd. Now $\gcd(n+3, n+1) \mid 2$ and $\gcd(n+3, n-1) \mid 4$; since all four numbers are odd, both gcds are 1. Similarly $\gcd(n-3, n-1) \mid 2$ and $\gcd(n-3, n+1) \mid 4$.

(3) We have $\gcd(n+2, n+1) = 1$ and $\gcd(n+2, n-1) \mid 3$. If $3 \mid n+2$ then $n \equiv 1$, so for $n \not\equiv 1 \pmod 3$ the second gcd is 1.

(4) We have $\gcd(n-2, n-1) = 1$ and $\gcd(n-2, n+1) \mid 3$, and $3 \nmid n-2$ when $n \not\equiv 2$. Also $2n-1 = 2(n-1)+1$ gives $\gcd(2n-1, n-1) = 1$. Finally $2(n+1) - (2n-1) = 3$ gives $\gcd(2n-1, n+1) \mid 3$, and $3 \mid 2n-1$ iff $n \equiv 2 \pmod 3$. □

Lemma 3.2 (moduli $2n \pm 1$)**.** *Let* $n \geq 1$*. The following sums are coprime to both* $2n-1$ *and* $2n+1$*:*

1. n *and* 4*, always; and* $2n-3$*, if* $n \geq 2$*;*
2. $n/2$*, if* n *is even;*
3. $n+1$*, and* $(n+1)/2$ *for odd* n*, if* $n \not\equiv 2 \pmod 3$*;*
4. $n-1$ *(for* $n \geq 2$*), and* $(n-1)/2$ *for odd* n*, if* $n \not\equiv 1 \pmod 3$.

Proof. Both moduli are odd.

(1) $\gcd(n, 2n \pm 1) \mid 1$. The number 4 is coprime to odd numbers. $\gcd(2n-3, 2n-1) \mid 2$ and $\gcd(2n-3, 2n+1) \mid 4$, and $2n-3$ is odd.

(2) With $m = n/2$: $\gcd(m, 4m \pm 1) \mid 1$.

(3) $\gcd(n+1, 2n+1) \mid 2(n+1) - (2n+1) = 1$. Also $\gcd(n+1, 2n-1) \mid 3$, and $3 \mid n+1$ iff $n \equiv 2$. For odd n put $x = (n+1)/2$, so $2n-1 = 4x-3$ and $2n+1 = 4x-1$. Then $\gcd(x, 4x-1) = 1$ and $\gcd(x, 4x-3) \mid 3$. Moreover $3 \mid x$ iff $n \equiv 5 \pmod 6$, which for odd n means $n \equiv 2 \pmod 3$.

(4) $\gcd(n-1, 2n-1) \mid (2n-1) - 2(n-1) = 1$. Also $\gcd(n-1, 2n+1) \mid 3$, and $3 \mid n-1$ iff $n \equiv 1$. For odd n put $y = (n-1)/2$, so $2n-1 = 4y+1$ and $2n+1 = 4y+3$. Then $\gcd(y, 4y+1) = 1$ and $\gcd(y, 4y+3) \mid 3$. Moreover $3 \mid y$ iff $n \equiv 1 \pmod 6$, which for odd n means $n \equiv 1 \pmod 3$. □

4 Proof of Theorem 1.2

4.1 The exceptional values $n = 2, 4$

Proposition 4.1. *There is no* $(n-1, n+1)$*-cycle of order* 2 *or of order* 4.

Proof. For $n = 2$ the only permutation with $p(0) = 0$ and $p(2) = 2$ is $(0, 1, 2)$. Its sum $1 + 2 = 3$ is not coprime to $n + 1 = 3$.

For $n = 4$ write the permutation as $(0, a, b, c, 4)$ with $\{a, b, c\} = \{1, 2, 3\}$; the moduli are 3 and 5. The sum $c + 4$ must avoid 5 and 6, so $c = 3$. Then $\{a, b\} = \{1, 2\}$, and $a + b = 3$ is not coprime to 3. □

4.2 Odd n (Sun's permutations)

The following two families are due to Sun [4, Remark 3.9(ii)]. We record them with their sums.

Construction 4.2 (Sun)**.** Let n be odd with $n \not\equiv 2 \pmod 3$. Define

$$p(i) = \begin{cases} i & i \text{ even},\ i \le n-1, \\ n-1-i & i \text{ odd}, \\ n & i = n. \end{cases}$$

This gives $(0,\, n-2,\, 2,\, n-4,\, 4,\, \ldots,\, 1,\, n-1,\, n)$. The sums are:

- $i + (n-2-i) = n-2$ for even $i < n-1$;
- $(n-1-i) + (i+1) = n$ for odd i;
- $s_{n-1} = (n-1) + n = 2n-1$ and $s_n = n$.

Construction 4.3 (Sun)**.** Let $n \ge 3$ be odd with $n \not\equiv 1 \pmod 3$. Define $p(0) = 0$, $p(n) = n$, $p(i) = i$ for odd i, and $p(i) = n+1-i$ for even $2 \le i \le n-1$. This gives $(0,\, 1,\, n-1,\, 3,\, n-3,\, \ldots,\, n-2,\, 2,\, n)$. The sums are:

- $s_0 = 1$;
- n for odd i;
- $n+2$ for even $2 \le i \le n-1$;
- $s_n = n$.

Both maps are permutations. In Construction 4.2 even positions carry even values and odd positions carry odd values; in Construction 4.3 even positions $i \ge 2$ carry the even values $n+1-i$. By Lemma 3.1(1),(3),(4) all sums are admissible. This proves Theorem 1.2 for odd n, including $n = 1$, where the permutation is $(0, 1)$ with sums $1, 1$.

4.3 Even n (new)

Let $n = 2m$ be even. Sun's zig-zag $i \mapsto i$ (i odd), $i \mapsto n+1-i$ (i even) has adjacent sums alternating between n and $n+2$. For even n it cannot end at $p(n) = n$, because it sends the even position n to the value 1. Our permutations therefore switch, in the middle, to the opposite phase $i \mapsto i$ (i even), $i \mapsto n+1-i$ (i odd), using a short gadget whose sums are $n \pm 3$, which are always admissible for even n (Lemma 3.1(2)).

Construction 4.4 ($n \equiv 0 \pmod 4$, $n \ge 8$)**.** Let $m = n/2$, which is even. Define

$$p(i) = \begin{cases} 0 & i = 0, \\ i & 1 \le i \le m-3,\ i \text{ odd}, \\ n+1-i & 1 \le i \le m-3,\ i \text{ even}, \\ m,\ m+3,\ m-1,\ m+1 & i = m-2,\ m-1,\ m,\ m+1 \text{ respectively}, \\ i & m+2 \le i \le n,\ i \text{ even}, \\ n+1-i & m+2 \le i \le n,\ i \text{ odd}. \end{cases}$$

That is, p is the sequence

$$0,\ 1,\ n-1,\ 3,\ n-3,\ \ldots,\ m+5,\ m-3,\ \big|\ m,\ m+3,\ m-1,\ m+1\ \big| \\ m+2,\ m-2,\ m+4,\ m-4,\ \ldots,\ n-2,\ 2,\ n.$$

Construction 4.5 ($n \equiv 2 \pmod 4$, $n \geq 10$)**.** Let $m = n/2$, which is odd. Define

$$p(i) = \begin{cases} 0 & i = 0, \\ i & 1 \leq i \leq m-1,\ i \text{ odd}, \\ n+1-i & 1 \leq i \leq m-1,\ i \text{ even}, \\ m+1,\ m-1,\ m+3,\ m,\ m-3 & i = m,\ m+1,\ m+2,\ m+3,\ m+4 \text{ respectively}, \\ i & m+5 \leq i \leq n,\ i \text{ even}, \\ n+1-i & m+5 \leq i \leq n,\ i \text{ odd}. \end{cases}$$

That is, p is the sequence

$$0,\ 1,\ n{-}1,\ 3,\ n{-}3,\ \ldots,\ m{-}2,\ m{+}2,\ \big|\ m{+}1,\ m{-}1,\ m{+}3,\ m,\ m{-}3\ \big|\ m{+}5,\ m{-}5,\ \ldots,\ n{-}2,\ 2,\ n.$$

Lemma 4.6. *Constructions 4.4 and 4.5 define permutations of $\{0, \ldots, n\}$ with $p(0) = 0$ and $p(n) = n$. All their cyclic adjacent sums lie in $\{1,\ n,\ n+2,\ n+3,\ n-3\}$. Precisely:*

- *Construction 4.4:*
  - $s_0 = 1$;
  - $s_i = n$ *(i odd) and* $s_i = n+2$ *(i even) for* $1 \leq i \leq m-4$;
  - $(s_{m-3}, s_{m-2}, s_{m-1}, s_m, s_{m+1}) = (n-3,\ n+3,\ n+2,\ n,\ n+3)$;
  - $s_i = n$ *(i even) and* $s_i = n+2$ *(i odd) for* $m+2 \leq i \leq n-1$;
  - $s_n = n$.
- *Construction 4.5:*
  - $s_0 = 1$;
  - $s_i = n$ *(i odd) and* $s_i = n+2$ *(i even) for* $1 \leq i \leq m-2$;
  - $(s_{m-1}, \ldots, s_{m+4}) = (n+3,\ n,\ n+2,\ n+3,\ n-3,\ n+2)$;
  - $s_i = n$ *(i even) and* $s_i = n+2$ *(i odd) for* $m+5 \leq i \leq n-1$;
  - $s_n = n$.

Proof. Permutation, Construction 4.4. Here m is even. We list the values taken by each block.

- First block, $1 \leq i \leq m-3$. The odd positions take the odd values $1, 3, \ldots, m-3$. The even positions $2 \leq i \leq m-4$ take the values $n+1-i$, which are the odd values $m+5, \ldots, n-1$.
- Gadget. It takes m and the odd values $m-1, m+1, m+3$.
- Last block, $m+2 \leq i \leq n$. The even positions take the even values $m+2, \ldots, n$. The odd positions $m+3 \leq i \leq n-1$ take the values $n+1-i$, which are the even values $2, \ldots, m-2$.

Together with $p(0) = 0$ every odd value in $[1, n-1]$ and every even value in $[0, n]$ occurs exactly once.

Permutation, Construction 4.5. Here m is odd and $m \geq 5$.

- First block, $1 \leq i \leq m-1$. It takes the odd values $1, \ldots, m-2$ and, at even positions, the odd values $n+1-i \in \{m+2, \ldots, n-1\}$.

- Gadget. It takes the odd value m and the even values $m-3, m-1, m+1, m+3$.
- Last block, $m+5 \le i \le n$. Here $m+5$ is even. It takes the even values $m+5, \dots, n$ and, at odd positions, the even values $n+1-i \in \{2, \dots, m-5\}$.

Again every value in $[0, n]$ occurs exactly once.

Sums. Inside a block of the first phase, an odd position i gives $i + (n+1-(i+1)) = n$ and an even position gives $(n+1-i)+(i+1) = n+2$. The same holds with the parities exchanged in a block of the second phase.

For Construction 4.4, the transitions around the gadget are

$$(m-3)+m = n-3, \quad m+(m+3) = n+3, \quad (m+3)+(m-1) = n+2,$$
$$(m-1)+(m+1) = n, \quad (m+1)+(m+2) = n+3.$$

For Construction 4.5 they are

$$(m+2)+(m+1) = n+3, \quad (m+1)+(m-1) = n, \quad (m-1)+(m+3) = n+2,$$
$$(m+3)+m = n+3, \quad m+(m-3) = n-3, \quad (m-3)+(m+5) = n+2.$$

Here $p(m-1) = n+1-(m-1) = m+2$, because $m-1$ is even. Finally $s_0 = 0+1 = 1$ and $s_n = n+0 = n$. □

Lemma 4.7 (mirror)**.** *If p is a permutation of $\{0, \dots, n\}$ with $p(0) = 0$ and $p(n) = n$, then so is $q(i) := n - p(n-i)$. Its adjacent sums are*

$$q(i) + q(i+1) = 2n - s_{n-1-i} \quad (0 \le i \le n-1), \qquad q(n) + q(0) = n.$$

Proof. The map $i \mapsto n-i$ reverses the order and the map $v \mapsto n-v$ reflects the values; both are involutions. We have $q(0) = n - p(n) = 0$ and $q(n) = n - p(0) = n$. The sum formula is immediate:

$$q(i) + q(i+1) = 2n - \big(p(n-i) + p(n-i-1)\big) = 2n - s_{n-1-i}.$$ □

Proposition 4.8. *For every even $n \ge 6$ there is an $(n-1, n+1)$-cycle of order n.*

Proof. For $n = 6$ take $(0, 1, 2, 4, 5, 3, 6)$. Its cyclic sums are $1, 3, 6, 9, 8, 9, 6$, all coprime to 5 and 7.

Let $n \ge 8$ be even, and let p be Construction 4.4 if $4 \mid n$ and Construction 4.5 otherwise.

- If $n \not\equiv 1 \pmod 3$, all sums of p lie in $\{1, n, n+2, n \pm 3\}$ by Lemma 4.6. These are admissible by Lemma 3.1(1)–(3).
- If $n \equiv 1 \pmod 3$, use the mirror q of p (Lemma 4.7). Its sums lie in $\{2n-1,\ n,\ n-2,\ n \mp 3\}$. These are admissible by Lemma 3.1(1),(2),(4), because $n \not\equiv 2 \pmod 3$. □

Proof of Theorem 1.2. Combine Proposition 4.1, Sun's odd constructions (Section 4.2) and Proposition 4.8. □

Example 4.9. The permutations produced by the proof for small even n are listed below. For $n = 10$ and $n = 16$ the mirror is used, because $n \equiv 1 \pmod 3$.

n	permutation $(p(0), \dots, p(n))$	cyclic sums $(s_0, \dots, s_n)$
6	$0, 1, 2, 4, 5, 3, 6$	$1, 3, 6, 9, 8, 9, 6$
8	$0, 1, 4, 7, 3, 5, 6, 2, 8$	$1, 5, 11, 10, 8, 11, 8, 10, 8$
10	$0, 8, 5, 2, 6, 4, 3, 7, 1, 9, 10$	$8, 13, 7, 8, 10, 7, 10, 8, 10, 19, 10$
12	$0, 1, 11, 3, 6, 9, 5, 7, 8, 4, 10, 2, 12$	$1, 12, 14, 9, 15, 14, 12, 15, 12, 14, 12, 14, 12$
14	$0, 1, 13, 3, 11, 5, 9, 8, 6, 10, 7, 4, 12, 2, 14$	$1, 14, 16, 14, 16, 14, 17, 14, 16, 17, 11, 16, 14, 16, 14$
16	$0, 14, 2, 12, 4, 10, 6, 7, 9, 5, 8, 11, 3, 13, 1, 15, 16$	$14, 16, 14, 16, 14, 16, 13, 16, 14, 13, 19, 14, 16, 14, 16, 31, 16$

Remark 4.10 (how the construction was found). Put $t = v - m \in [-m, m]$. A sum $v + w = n + c$ becomes $t + t' = c$. The always-admissible sums n and $n \pm 3$ correspond to the reflections $t \mapsto -t$ and $t \mapsto \pm 3 - t$. Adding $n+2$, which corresponds to $t \mapsto 2 - t$, makes the two zig-zags $0, 2, -2, 4, -4, \ldots$ and $1, -1, 3, -3, \ldots$ available. They are glued near $t = 0$ by the gadget $2, 1, -1, 3, 0, -3, 5$, and closed near $t = \pm m$ by the sum $v + w = 1$. The mirror $t \mapsto -t$ exchanges $n + 2$ with $n - 2$, which covers the residue class $n \equiv 1 \pmod 3$. An exhaustive search shows that for $n = 8$ there is no admissible cycle using only the sums n, $n \pm 3$ and powers of two, so these universally admissible sums alone do not suffice.

5 Proof of Theorem 1.3

Now the moduli are $2n - 1$ and $2n + 1$.

Construction 5.1 ($n \not\equiv 2 \pmod 3$, n even)**.** Let $m = n/2$, $p(0) = 0$ and $p(n) = n$. For $1 \le i \le n - 1$ put $p(i) = m - \frac{i-1}{2}$ for odd i and $p(i) = m + \frac{i}{2}$ for even i. This gives

$$(0,\ m,\ m+1,\ m-1,\ m+2,\ m-2,\ \ldots,\ n-1,\ 1,\ n).$$

The odd positions take $\{1, \ldots, m\}$ and the even positions $2 \le i \le n - 2$ take $\{m + 1, \ldots, n - 1\}$. The sums are:

- $s_0 = m$;
- $s_i = n + 1$ for odd $i \le n - 3$ and $s_i = n$ for even $2 \le i \le n - 2$;
- $s_{n-1} = 1 + n$ and $s_n = n$.

Construction 5.2 ($n \not\equiv 2 \pmod 3$, n odd)**.** Let $m = (n - 1)/2$, $p(0) = 0$ and $p(n) = n$. For $1 \le i \le n - 1$ put $p(i) = m + 1 + \frac{i-1}{2}$ for odd i and $p(i) = m + 1 - \frac{i}{2}$ for even i. This gives

$$(0,\ m+1,\ m,\ m+2,\ m-1,\ \ldots,\ n-1,\ 1,\ n).$$

The odd positions take $\{m + 1, \ldots, n - 1\}$ and the even positions take $\{1, \ldots, m\}$. The sums are:

- $s_0 = (n + 1)/2$;
- $s_i = n$ for odd i and $s_i = n + 1$ for even $2 \le i \le n - 3$;
- $s_{n-1} = n + 1$ and $s_n = n$.

For $n = 1$ this is $(0, 1)$.

Construction 5.3 ($n \equiv 2 \pmod 3$, $n \ge 8$)**.** Let $m = \lfloor n/2 \rfloor$ and $p(0) = 0$. For $1 \le i \le n - 6$ put:

- if n is even: $p(i) = m + \frac{i-1}{2}$ for odd i, and $p(i) = m - \frac{i}{2}$ for even i;
- if n is odd: $p(i) = m - \frac{i-1}{2}$ for odd i, and $p(i) = m + \frac{i}{2}$ for even i.

Finally put $(p(n-5), p(n-4), \ldots, p(n)) = (1,\ n-1,\ n-2,\ 2,\ n-3,\ n)$. So p is

$$0,\ \underbrace{m,\ m \mp 1,\ m \pm 1,\ \ldots,\ 4,\ n-4,\ 3}_{\text{positions } 1,\ldots,n-6},\ 1,\ n-1,\ n-2,\ 2,\ n-3,\ n,$$

with the upper signs for even n. The zig-zag block takes exactly the values $\{3, \ldots, n - 4\}$. The sums are:

- $s_0 = m$;
- along the zig-zag, $n-1$ and n alternately;
- $s_{n-6} = 3+1 = 4$;
- then $n,\ 2n-3,\ n,\ n-1,\ 2n-3$;
- $s_n = n$.

Verification of Construction 5.3. n even. The odd positions $1, 3, \ldots, n-7$ take the values $m, m+1, \ldots, m+\frac{n-8}{2} = n-4$. The even positions $2, \ldots, n-6$ take the values $m-1, \ldots, m-\frac{n-6}{2} = 3$. An odd position $i = 2k+1$ gives $(m+k)+(m-k-1) = n-1$, and an even position $i = 2k$ gives $(m-k)+(m+k) = n$.

n odd. Here $n = 2m+1$. The odd positions take $m, m-1, \ldots, 3$ and the even positions take $m+1, \ldots, n-4$. The zig-zag sums are $(m-k)+(m+k+1) = n$ and $(m+k)+(m-k) = n-1$.

In both cases the last zig-zag value is $p(n-6) = 3$, and the tail gives the sums listed. □

Proof of Theorem 1.3. For $n = 2$ the only candidate $(0, 1, 2)$ has the sum $3 = 2n-1$, so no such permutation exists.

For $n = 5$ take $(0, 1, 4, 3, 2, 5)$. Its sums $1, 5, 7, 5, 7, 5$ are coprime to 9 and 11.

For all other n we use the constructions above.

- If $n \not\equiv 2 \pmod 3$, use Construction 5.1 or Construction 5.2. All sums lie in $\{n/2,\ n,\ n+1\}$ or $\{(n+1)/2,\ n,\ n+1\}$, which are admissible by Lemma 3.2(1)–(3).
- If $n \equiv 2 \pmod 3$ and $n \geq 8$, use Construction 5.3. All sums lie in $\{m,\ n-1,\ n,\ 4,\ 2n-3\}$, which are admissible by Lemma 3.2(1),(2),(4). Here, for odd n, $m = (n-1)/2$ and $n \not\equiv 1 \pmod 3$. □

Example 5.4. For $n = 4$, which the arXiv formulation of the companion conjecture excludes, Construction 5.1 gives $(0, 2, 3, 1, 4)$ with sums $2, 5, 4, 5, 4$ (moduli $7, 9$). Further examples:

- $n = 8$: $(0, 4, 3, 1, 7, 6, 2, 5, 8)$, with sums $4, 7, 4, 8, 13, 8, 7, 13, 8$ (moduli $15, 17$);
- $n = 11$: $(0, 5, 6, 4, 7, 3, 1, 10, 9, 2, 8, 11)$, with sums $5, 11, 10, 11, 10, 4, 11, 19, 11, 10, 19, 11$ (moduli $21, 23$).

6 Formal verification

All results above are verified in a single self-contained Lean 4 [12] file, `lean/Main.lean` (about 830 lines). It uses Lean version 4.34.1 with the core library only: no Mathlib and no external dependencies. It contains no `sorry`, `axiom`, `admit` or `native_decide`.

Statement. The two main theorems are stated as follows. Their definitions appear directly above them in the file and are reproduced here verbatim.

```
/-- The cyclic adjacent sums of `l = [a_0, a_1, ..., a_k]`, namely
`[a_0 + a_1, a_1 + a_2, ..., a_{k-1} + a_k, a_k + a_0]` (entry `i` is `l[i] + l[(i+1)
    mod length]`). -/
def cyclicSums (l : List Nat) : List Nat :=
  (List.range l.length).map (fun i => l.getD i 0 + l.getD ((i + 1) % l.length) 0)
```

```
/-- 'SunCycle n A B l': the list 'l = [i_0, i_1, ..., i_n]' is a permutation of '[0, 1,
    ..., n]'
with 'i_0 = 0' and 'i_n = n', and every cyclic adjacent sum 'i_0+i_1, ..., i_{n-1}+i_n,
    i_n+i_0'
is coprime to both 'A' and 'B'.  A228886(n) counts 'SunCycle n (n - 1) (n + 1)'. -/
def SunCycle (n A B : Nat) (l : List Nat) : Prop :=
  l.Perm (List.range (n + 1)) ∧ l.head? = some 0 ∧ l.getLast? = some n ∧
  ∀ s ∈ cyclicSums l, Nat.gcd s A = 1 ∧ Nat.gcd s B = 1
```

```
/-- **Sun's conjecture (OEIS A228886).**  For every 'n ≥ 1' there is a permutation
'l = [i_0, ..., i_n]' of '[0, ..., n]' with 'i_0 = 0', 'i_n = n' such that all 'n + 1'
    cyclic adjacent
sums 'i_0+i_1, ..., i_{n-1}+i_n, i_n+i_0' are coprime to both 'n - 1' and 'n + 1',
if and only if 'n ≠ 2' and 'n ≠ 4'.   (I.e. A228886(n) > 0 for all n ≥ 1 except n = 2,
    4.) -/
theorem sun_conjecture (n : Nat) (hn : 1 ≤ n) :
    (∃ l, SunCycle n (n - 1) (n + 1) l) ↔ (n ≠ 2 ∧ n ≠ 4) := by
  rw [exists_list_iff]; exact sun_conjecture_fun n hn

/-- **Sun's companion conjecture** (A228886 comments; arXiv:1309.1679 Remark 3.9(i)):
the same with the moduli '2n - 1' and '2n + 1'; it holds for every 'n ≥ 1' except 'n =
    2'. -/
theorem sun_companion (n : Nat) (hn : 1 ≤ n) :
    (∃ l, SunCycle n (2 * n - 1) (2 * n + 1) l) ↔ n ≠ 2 := by
  rw [exists_list_iff]; exact sun_companion_fun n hn
```

Axioms and resources. `#print axioms` reports:

```
'A228886.sun_conjecture' depends on axioms: [propext, Classical.choice, Quot.sound]
'A228886.sun_companion' depends on axioms: [propext, Classical.choice, Quot.sound]
```

These are the standard axioms of Lean's foundations.

- The check takes about 480 s of wall-clock time (about 352 s of CPU time) with a peak memory of about 1.6 GB, on a shared machine.
- The file sets `set_option Elab.async false`, because asynchronous elaboration failed to create threads under that machine's resource limits. The option has no logical effect.

Statement audit.

- `List.range (n + 1)` is $[0, 1, \ldots, n]$. So `l.Perm (List.range (n + 1))` says that l lists $0, \ldots, n$, each exactly once.
- `l.head? = some 0` and `l.getLast? = some n` say $i_0 = 0$ and $i_n = n$.
- For a list of length $n+1$, `cyclicSums l` is the list of $l[i]+l[(i+1) \bmod (n+1)]$ for $i = 0, \ldots, n$, that is $i_0 + i_1, \ldots, i_{n-1} + i_n, i_n + i_0$. Both indices are always smaller than the length, so the default value 0 of `getD` is never used.

- `Nat.gcd s A = 1` is coprimality. Under the hypothesis $1 \le n$ the natural-number subtractions $n-1$ and $2n-1$ are exact. For $n=1$, $\gcd(s,0)=s$, matching the OEIS value $a(1)=1$.
- The theorems are equivalences for every $n \ge 1$, which is the offset of A228886. They include the non-existence statements at $n=2,4$ (respectively $n=2$).
- The file also contains `decide`-checked examples. They show that a correct permutation is accepted, and that a list violating the gcd condition, a rotated cycle (same cyclic sums but $i_0 \neq 0$), and a non-permutation are all rejected.

Proof structure in Lean.

- An equivalent function form `SunFun n A B p` is used, in which $p : \mathbb{N} \to \mathbb{N}$ is injective on $\{0,\dots,n\}$ with values in $\{0,\dots,n\}$. The lemma `exists_list_iff` transfers between the list and function forms, using a pigeonhole lemma `surj_of_inj` proved by induction.
- Each construction of Sections 4 and 5 is a Lean function defined by a cascade of `if`s: `oddA`, `oddB`, `evenA`, `evenB`, the mirror, `compA`, `compB`, `compC`, and the explicit cases $n=5,6$.
- Bounds, injectivity and the list of adjacent sums are proved by splitting all branches and calling `omega`; injectivity is checked over all pairs of branches.
- The admissibility lemmas of Section 3 are reduced, by Euclid-type rewriting, to $\gcd(r,k)=1$ with $k \le 4$, which is settled by `decide`.
- The impossibility at $n=2,4$ is a short case analysis.

Independent numerical check. The script `data/verify_constructions.py` transcribes the Lean definitions into Python. It confirms every construction for $n \le 3000$ and recomputes A228886(1..11) by brute force, in agreement with the OEIS data. This check is not part of the proof.

7 Limitations and open questions

- **Novelty.** The claim of novelty rests on a literature and web search, as described in Section 1. In particular we could not check Sun's 2021 book [6].
- **Depth.** The proofs are elementary. They give existence only and say nothing about the size of A228886(n), which grows irregularly (for instance $a(11)=34$ while $a(12)=4021156$). Its dependence on the factorisation of n^2-1 is not addressed.
- **Squares.** Sun's variant A229130 [3] uses $i_k^2 + i_{k+1}$ in place of $i_k + i_{k+1}$. It conjectures existence for all $n \notin \{2,5,11\}$, and more generally that the exceptional sets $E(k)$ for $i_k^k + i_{k+1}$ are finite. It remains open as far as we know. The linear zig-zag method used here does not apply directly.
- **Other problems in [4].** Other circular-permutation problems there involve primes (twin primes, Sophie Germain primes, prime circles) or primitive roots. They are of a different nature, and most remain open.

References

[1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A228886 (Z.-W. Sun, Sep 07 2013; terms to $a(26)$ by M. Alekseyev, Jul 07 2026). https://oeis.org/A228886.

[2] OEIS Foundation Inc., sequence A228860 (Z.-W. Sun, Sep 05 2013). https://oeis.org/A228860.

[3] OEIS Foundation Inc., sequence A229130 (Z.-W. Sun, Sep 15 2013). https://oeis.org/A229130.

[4] Z.-W. Sun, *Some new problems in additive combinatorics*, Nanjing Univ. J. Math. Biquarterly 36 (2019), 134–155; arXiv:1309.1679 [math.NT], v9 (March 2, 2020) consulted. https://arxiv.org/abs/1309.1679.

[5] Z.-W. Sun, *Introduction to Combinatorial Number Theory (VI): Permutations, Determinants and Permanents*, on-line talk slides, June 26, 2020. http://maths.nju.edu.cn/~zwsun/CNT-talk6.pdf.

[6] Z.-W. Sun, *New Conjectures in Number Theory and Combinatorics* (in Chinese), Harbin Institute of Technology Press, 2021. (Not consulted for this work.)

[7] Y.-F. She and X.-Q. Luo, *Proof of a conjecture on permutations*, arXiv:2606.18311, 2026.

[8] Y.-F. She, *Hamiltonicity in graphs defined by primes and primitive elements*, arXiv:2609.19114, 2026.

[9] Google DeepMind, *formal-conjectures* (OEIS folder), https://github.com/google-deepmind/formal-conjectures, accessed October 8, 2026.

[10] G. Tsoukalas, A. Kovsharov, S. Shirobokov, A. Surina, M. Firsching, G. Bérczi, F. J. R. Ruiz, A. Suggala, A. Z. Wagner, E. Wieser, L. Yu, A. Huang, et al., *Advancing mathematics research with AI-driven formal proof search*, arXiv:2605.22763, 2026.

[11] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941, 2026.

[12] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in Automated Deduction – CADE 28, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.